SUSHMA NIRESH

Philadelphia, PA | +1 602-807-8786 | sniresh@asu.edu | linkedin.com/in/sushma-n | github.com/Sushmaniresh

EDUCATION

M.S. in Data Science, Analytics & Engineering (Electrical Engineering concentration) May 2026
Arizona State University
Relevant coursework: Biomedical Signal Processing & Machine Learning, Image & Speech Processing, Generative AI: Theory & Practice

B.E. in Electronics & Communication May 2021
JSS Science & Technology University (SJCE), Mysuru, India

PROFESSIONAL EXPERIENCE

Research Assistant, Banerjee Lab, ASU–Mayo Clinic Arizona (NIH-funded) Jan 2026 – Oct 2026

- Fine-tuned an ImageNet-pretrained, two-head MobileNetV2 in PyTorch to read pH and lactate from a microfluidic sensor for on-device iOS inference: **99.3%** accuracy on both on a held-out set; augmentation raised both-correct accuracy from **90.7%** to **98.5%**.
- Built a **1,792**-image dataset; after real-world underperformance, worked with the PI to identify collector bias and retrained on multi-collector data.
- Evaluated beyond accuracy with Grad-CAM++ and a perturbation study, finding weakness at intermediate values that real-world app testing confirmed (**100%** vs **48%**).

AI Engineer Co-op, Tarifflo Inc. (YC-backed trade compliance), Provo, UT May 2025 – Jan 2026

- Proposed and built a production Vertex AI Gemini pipeline that turns tariff legal text (code ranges, exceptions, country exclusions) into schema-constrained JSON for a live duty-rate product, replacing a regex parser that broke on nearly every revision.
- Designed context management and retrieval after whole-document prompts overflowed the context window: each call retrieves only the subchapter text and tariff history for one code.
- Built the evaluation layer: regression tests from **8** real failure modes (truncation, dropped delimiters, hallucinated codes) plus an automated diff of duty rates against an independent source, with **zero unrecovered incidents** across **23** regulatory revisions.
- Owned a customer-reported production incident end to end: migrated the pipeline to Gemini 2.0 Flash when 1.5 was retired, adapted to its structured-output changes, and backfilled missed updates.
- Built a BigQuery ETL for **500K+** records with SCD2 versioning, data-quality gates, and nightly revision-triggered runs; cut query latency by **50%** (40s to 20s).

Associate Software Engineer, Maveric Systems Ltd. (Client: Citi Bank), Bengaluru, India Feb 2021 – Jul 2024

- Worked directly with Citi stakeholders to gather requirements, resolve data discrepancies, and lead UAT for regulatory data-migration releases.
- Architected a reusable Python validation framework for **4–40 GB** financial data transfers across **7** regions, integrated into CI/CD (Jenkins, Bitbucket).
- Built diagnostics tooling for data-discrepancy investigation and a FinBERT news-sentiment proof of concept.

PROJECTS

- **Multimodal Chest X-ray Analysis & Report Generation** (ASU final project, team): multi-view ConvNeXt classifier for **14** findings on **44K** MIMIC-CXR studies with patient-wise splits and per-label thresholds (**0.71** micro-F1 on holdout), plus retrieval-based report generation.
- **ECG Arrhythmia Detection** (Biomedical Signal Processing & ML course): filtered and segmented MIT-BIH ECG (band-pass, R-peak detection); classified arrhythmias with ML and CNN/BiLSTM models.
- **Diabetic Retinopathy Detection** (Data Science Open Lab, ASU): CNN classifier for 5-level severity grading on APTOS 2019 retinal fundus images (TensorFlow/Keras), with background removal and data augmentation.
- **RAG Chunking & Retrieval Comparison** (ASU CSE 511, team): built the late-chunking and retrieval modules, FAISS index, and Streamlit UI comparing four chunking strategies.

TECHNICAL SKILLS

- **AI/ML:** PyTorch, Hugging Face, fine-tuning, computer vision, NLP, model evaluation, on-device inference
- **LLMs:** Vertex AI (Gemini), OpenAI & Anthropic APIs, RAG, FAISS, embeddings, context management, structured outputs (Pydantic), LLM evaluation
- **MLOps & Data:** Python, SQL, GCP (Vertex AI, BigQuery), AWS, Docker, CI/CD (Jenkins), Git, FastAPI, PostgreSQL, ETL
- **Certifications:** NLP & ML Specializations (DeepLearning.AI), Intro to Generative AI (Google Cloud)